

Chapter 5 Regression-Based Classification

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Decision Trees

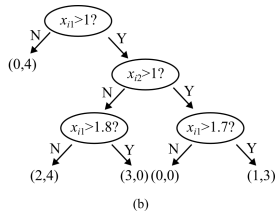
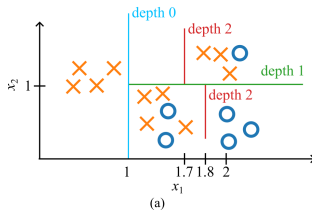
Decision Trees are flexible models used for:

- Classification
- Regression
- Multi-output prediction

Key ideas:

- Split data using simple rules
- Build a tree of decisions
- Make predictions by following a path

They are also the building blocks of Random Forests.



Basic concept of a decision tree.

Decision Trees

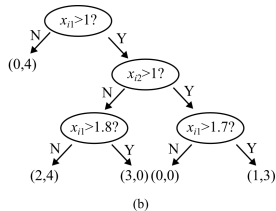
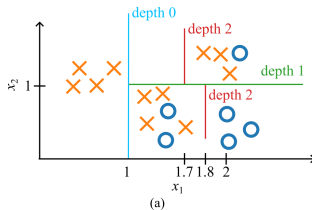
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Decision Tree Classification

A decision tree partitions the feature space into regions.

For the Iris dataset:

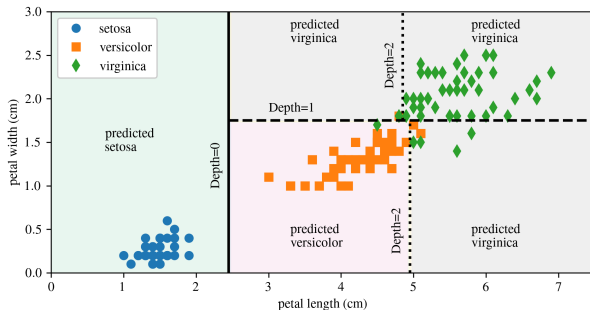
► Root split:

$$\text{petal length} \leq 2.45$$

► Second split:

$$\text{petal width} \leq 1.75$$

Each region is assigned a predicted class.



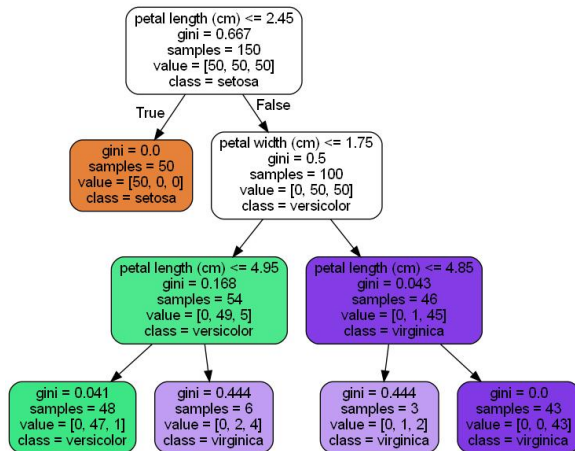
Decision tree decision boundaries.

Reading a Decision Tree

To classify a new sample:

1. Start at the root node
2. Answer the split question
3. Move left or right
4. Continue until a leaf node is reached

The leaf node gives the predicted class.



Decision tree trained on the Iris dataset.

Node Information

Each node in a classification tree stores useful information.

- ▶ **class**: predicted class at the node
- ▶ **samples**: number of training instances that reach the node
- ▶ **value**: class counts at the node
- ▶ **gini**: impurity of the node

A node is pure when all samples belong to the same class:

$$G_i = 0$$

Darker node colors indicate purer predictions.

Gini Impurity

Gini impurity measures how mixed the classes are at a node.

$$G_i = 1 - \sum_{k=1}^n p_{i,k}^2$$

where:

- ▶ $p_{i,k}$ is the proportion of class k at node i
- ▶ $G_i = 0$ means the node is pure
- ▶ Larger values indicate more mixed classes

Example:

$$1 - \left(\frac{0}{54}\right)^2 - \left(\frac{49}{54}\right)^2 - \left(\frac{5}{54}\right)^2 \approx 0.168$$

Class Probability Estimation

Decision Trees can estimate class probabilities.

Procedure:

1. Follow the sample to a leaf node
2. Count the training samples in that leaf
3. Compute the fraction belonging to each class

Example leaf node:

$$\text{value} = [0, 49, 5]$$

Probabilities:

$$P(\text{setosa}) = 0\%$$

$$P(\text{versicolor}) = \frac{49}{54} = 90.7\%$$

Example 7.1: Decision Tree Classifier

- ▶ Use `DecisionTreeClassifier`
- ▶ Classify Iris species using petal features
- ▶ Train a tree with a limited depth
- ▶ Visualize the learned decision rules with Graphviz

The exported tree shows:

- ▶ Feature thresholds
- ▶ Gini impurity
- ▶ Sample counts
- ▶ Class distributions

Graphviz viewer:

<https://dreampuf.github.io/GraphvizOnline/>